

Bruk av kunstig intelligens i matematikkundervisning

En kasssstudie av Newtons metode og halveringsmetoden i et flipped-classroom-opplegg

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Abstrakt. Denne kasssstudien undersøker hvordan kunstig intelligens (KI) kan anvendes som veileder og læringspartner i matematikkundervisning innen et flipped-classroom-rammeverk. Studentene, som ikke tidligere har arbeidet med Newtons metode, halveringsmetoden eller Python-programmering, får i oppgave å utforske disse temaene selvstendig ved hjelp av KI før felles gjennomgang i klassen. Studien fokuserer på *prompt engineering* som metode for å etablere en faglig produktiv dialog med KI, og på sammenligningen av Newtons metode og halveringsmetoden for å finne nullpunktet til $f(x) = x^2 - \cos(x)$. Gjennom en kombinasjon av teoretisk redegjørelse, Python-implementasjon og strukturerte refleksjonsspørsmål undersøkes konvergenshastighet, valg av startverdi og toleranse, samt metodens styrker og svakheter. Resultatene indikerer at KI kan fremme dybdelæring og begrepsforståelse når den brukes med presise og faglig funderte instruksjoner, men at læringsutbyttet i stor grad avhenger av studentens evne til å stille gode spørsmål og kritisk vurdere KI-genererte svar.

Emneord: kunstig intelligens, matematikdidaktikk, flipped classroom, prompt engineering, numeriske metoder, Newtons metode, halveringsmetoden.

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1. Innledning

I denne oppgaven anvendes prinsippet om *omvendt undervisning* (flipped classroom), der studentene selv utforsker et nytt tema ved hjelp av digitale ressurser og kunstig intelligens (KI) før temaet gjennomgås i fellesskap i klassen. Studentene har ikke tidligere lært Newtons metode eller halveringsmetoden, og heller ikke programmert grundig i Python. Hensikten er derfor å bruke KI som en *veileder* og *læringspartner* for å oppdage og forstå nye konsepter, og samtidig trene ferdigheter i å stille gode spørsmål, innhente forklaringer og i samarbeid med KI løse en oppgave innen numeriske metoder.

2. Hensikt med oppgaven

Hensikt

Formålet med denne oppgaven er å lære studenten hvordan KI (for eksempel ChatGPT) kan brukes som **veileder**, **studievenn** og **samarbeidspartner**. Gjennom dialog med KI skal studenten erfare hvordan en **god samtale** kan føre til bedre forståelse, tydeligere tenkning og bedre resultater i matematikken.

3. Prompt engineering og den gode samtalen

Om prompt engineering

«Prompt engineering» handler om å stille gode spørsmål og formulere forespørsler på en måte som hjelper KI til å gi nyttige og presise svar. Den gode samtalen med KI oppstår når man:

- stiller spørsmål som viser hva man vil forstå, ikke bare hva man vil ha
- ber om forklaring, ikke bare løsning
- følger opp med refleksjon og presisering

3.1 Eksempler på prompt-kvalitet

Dårlig prompt:

«Gi meg koden for Newtons metode i Python.»

Denne prompten er for kort og gir ingen kontekst. KI vet ikke hva slags funksjon man arbeider med, hvilket formål man har, eller hvilket nøyaktighetsnivå man ønsker.

God prompt:

«Jeg skal sammenligne Newtons metode og halveringsmetoden for å finne nullpunktet til $x^2 - \cos(x) = 0$. Kan du hjelpe meg å skrive Python-kode som beregner løsningen, teller antall iterasjoner for hver metode, og forklarer hvorfor Newtons metode konvergerer raskere?»

Denne prompten er konkret, faglig og inviterer til forklaring. Den hjelper KI til å gi både kode og forståelse.

4. Kasus: Sammenligning av Newtons metode og halveringsmetoden

4.1 Problemstilling og hensikt

Caset ble utformet for å trene studentene i å forstå og implementere halveringsmetoden og Newtons metode for å løse ligningen

$$f(x) = x^2 - \cos(x) = 0. \quad (1)$$

Studentene får følgende oppgave:

«Bruk KI som støtte til å forklare og implementere halveringsmetoden og Newtons metode i Python. Sammenlign løsningene, antall iterasjoner og hvor raskt metodene konvergerer. Diskuter hvilken metode som er mest effektiv, og hvorfor.»

4.2 Arbeidskrav

Studentene skal bruke KI (for eksempel ChatGPT) til å:

- a) Forklare hva Newtons metode og halveringsmetoden er.
- b) Si litt om skjæringssetningen, og forklare hvorfor startverdi skal velges i et intervall $[a, b]$ der $f(a) \cdot f(b) < 0$ (hint: ligningen har minst én løsning hvis funksjonsverdiene i endepunktene har motsatt fortegn).

- c) Utvikle Python-kode som
- setter opp funksjonen $f(x)$
 - tar startverdi og toleranse som input
 - implementerer Newtons metode og halveringsmetoden som funksjoner
 - definerer toleranser (estimeringsavvik) $e = 10^{-2}$, $e = 10^{-4}$ og $e = 10^{-6}$
 - bestemmer løsningene med begge metodene
 - skriver ut løsningen med begge metodene med tilhørende toleranser og antall iterasjoner for sammenligning
- d) Reflektere over styrker og svakheter ved metodene.

4.3 Refleksjonsspørsmål

- a) Hva må endres i programmet for å få bedre tilnærming eller raskere konvergens?
- b) Hva må endres i programmet for å få raskere konvergens?
- c) Hvordan kan man velge en passende startverdi?
- d) Forklar hvorfor intervallet $[0, 1]$ er et godt utgangspunkt for halveringsmetoden for å finne nullpunktet til $f(x) = x^2 - \cos(x)$ (hint: undersøk om $f(0) \cdot f(1) < 0$, husk å sette kalkulatoren på radianer). Kan $x_0 = 0,5$ være en passende startverdi for Newtons metode?
- e) Hvordan påvirker valg av startverdi og toleranse resultatene?
- f) Hvilken metode gir best balanse mellom presisjon og stabilitet?
- g) Hvilken metode gir svaret med færrest iterasjoner?

4.4 Kort oppsummering av kasusstudien

Studentene bes avslutningsvis svare kort på følgende fem spørsmål:

1. Hva synes du om flipped classroom (omvendt undervisning) – altså at man lærer seg grunnleggende stoff på egen hånd før temaet blir undervist i timen?
2. Hva tenker du om å bruke kunstig intelligens (KI) som støtte for å lære nye ting og som veileder i læringsprosessen?
3. Hvor viktig mener du det er å kunne sette opp og bruke gode KI-agenter for å lære seg nye temaer og løse oppgaver?
4. I hvilken grad mener du at KI kan fungere som en god veileder for studenter – for eksempel ved å hjelpe dem å finne ny kunnskap, forstå begreper og teorier, eller øve på regneoppgaver i matematikk?
5. Tror du KI kan bidra til dybdelæring og gi bedre oversikt over detaljer i et tema – forutsatt at man gir agentene gode instruksjoner?

«Den gode samtalen med KI skaper refleksjon – ikke bare løsninger.»

5. Python-implementasjon

På eksamen i MAT110 får man ikke oppgave i å programmere, men man må kunne forklare hva et gitt program utfører. Følgende program implementerer halveringsmetoden og Newtons metode for ligningen (1) og sammenligner metodene for tre toleranser.

```
1 import math
2
3 def f(x):
4     return x**2 - math.cos(x)
5
6 def halvering(a, b, tol=1e-3):
7     n = 0
8     while (b - a)/2 > tol:
9         c = (a + b)/2
10        if f(c) == 0:
11            break
12        elif f(a)*f(c) < 0:
13            b = c
14        else:
15            a = c
16        n += 1
17    return c, n
18
19 def newton(x0, tol=1e-3):
20     n = 0
21     while abs(f(x0)) > tol:
22         df = 2*x0 + math.sin(x0)
23         x0 = x0 - f(x0)/df
24         n += 1
25     return x0, n
26
27 # Definer toleranser
28 toleranser = [1e-3, 1e-4, 1e-6]
29
30 # Print tabell-header
31 print(f"{'Tol':>10}|{'n_halv':>8}|{'x_halv':>10}|"
32       f"{'n_newt':>8}|{'x_newt':>10}")
33 print("-"*60)
34
35 for tol in toleranser:
36     Losn_h, n_h = halvering(0, 1, tol)
37     Losn_n, n_n = newton(0.5, tol)
38     print(f"{tol:10.1e}|{n_h:8d}|{Losn_h:10.6f}|"
39           f"{n_n:8d}|{Losn_n:10.6f}")
```

Listing 1: Python-implementasjon av halveringsmetoden og Newtons metode for $f(x) = x^2 - \cos(x)$.

6. Refleksjon og avsluttende ord

Kasusstudien illustrerer flere sentrale aspekter ved bruk av KI i matematikkundervisning. For det første viser sammenligningen av Newtons metode og halveringsmetoden at de to metodene har komplementære egenskaper: Newtons metode konvergerer kvadratisk og krever derfor vesentlig færre iterasjoner for å oppnå en gitt toleranse, men forutsetter at den deriverte eksisterer og er veldefinert i nærheten av nullpunktet, og at startverdien ligger tilstrekkelig nær løsningen. Halveringsmetoden på sin side konvergerer lineært og er derfor langsommere, men er robust så

lenge f er kontinuerlig og skifter fortegn på intervallet $[a, b]$ ifølge skjæringssetningen. Dette gjør halveringsmetoden til et trygt valg når man ikke har god kjennskap til funksjonens oppførsel, mens Newtons metode er å foretrekke når man har en god startverdi og ønsker høy presisjon raskt.

For det andre demonstrerer opplegget hvordan KI kan fungere som en *katalysator* for læring snarere enn som en erstatning for egen tenkning. Når studenten formulerer presise og faglig funderte spørsmål, gir KI svar som kan brukes til å bygge begrepsforståelse, utforske alternative fremgangsmåter og avdekke misforståelser. Den gode samtalen kjennetegnes ved at studenten ikke bare mottar en løsning, men også får innsikt i *hvorfor* løsningen er gyldig, *hvilke* antagelser som ligger til grunn, og *hvordan* metoden kan generaliseres til andre problemer.

For det tredje peker kasusstudien på noen didaktiske utfordringer. KI kan gi overbevisende, men feilaktige svar, og studenten må derfor utvikle kritisk vurderingsevne. Videre kan en for ensidig avhengighet av KI føre til at grunnleggende ferdigheter, som å utlede den deriverte analytisk eller å vurdere konvergensforhold, blir underutviklet. Flipped classroom-rammeverket avhjelper dette ved at den felles gjennomgangen i klassen gir anledning til å kvalitetssikre og utdype det studentene har lært på egen hånd.

Avslutningsvis indikerer erfaringene fra denne kasusstudien at KI kan bidra til dybdelæring og gi bedre oversikt over detaljer i et tema, forutsatt at studenten gis gode instruksjoner og oppmuntres til å stille spørsmål som fremmer forståelse. Den gode samtalen med KI skaper refleksjon – ikke bare løsninger – og det er nettopp denne refleksjonen som er målet med undervisningsopplegget.

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Artificial Intelligence as a Learning Partner in Mathematics Education: Teacher and Student Perspectives

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Abstract—The rapid integration of generative AI into higher education has fundamentally changed mathematics teaching and learning. Based on experiences at HVL and NTNU in Ålesund, and informed by recent literature, this paper presents a dual perspective: how teachers can leverage AI to design differentiated instruction, reflective tasks, and assessment; and how students can learn to use AI as a genuine learning partner rather than a shortcut. We identify six major pitfalls of uncritical AI use and propose pedagogical countermeasures. A concrete flipped-classroom case study (Newton’s method vs. bisection) illustrates how AI can be integrated as a reflective guide, using structured reflection questions and self-assessment. Preliminary findings indicate that teachers view generative AI as an opportunity to enrich differentiated teaching, while students need explicit training in prompt engineering and self-regulation to avoid becoming passive consumers. Student feedback confirms that reflection and self-assessment have increased their awareness of learning processes, not only in mathematics but across other subjects.

Index Terms—Artificial intelligence, mathematics education, prompt engineering, flipped classroom, reflective learning, metacognition.

I. INTRODUCTION

In just a few years, artificial intelligence has moved from research laboratories into the daily workflow of students and lecturers. At HVL and NTNU in Ålesund, we have observed that AI tools such as ChatGPT are now used by the majority of students for proofreading, structuring arguments, solving mathematical problems, and even generating code. This shift has happened faster than our educational systems could adapt [1].

However, attendance in exercise sessions has declined, and students increasingly rely on AI for guidance rather than engaging with teachers or peers. To counter this, we must design pedagogical activities that teach students how to use AI as a reflective tool, and that help them become aware of their own learning processes. This paper presents a concrete approach to achieve this.

For lecturers, AI is transforming our opportunities while also presenting new challenges. On one hand, we can now

generate personalised problem sets, analyse student misconceptions at scale, and design flipped classroom activities that were previously too time-consuming. On the other hand, traditional take-home assignments are no longer viable because AI can solve them instantly. Recent research shows that GenAI can attain performance comparable to first-class degree level in first-year undergraduate mathematics examinations, highlighting the urgent need for assessment redesign [4].

Today’s students differ from previous cohorts. They often have less theoretical background, struggle with abstract thinking, and are accustomed to learning through problem-solving and algorithmic thinking rather than theoretical exposition. Traditional one-way communication in lectures is becoming outdated. As a result, the lecturer’s role is changing from a transmitter of knowledge to a facilitator and guide. This paper presents a pedagogical approach for helping students take ownership of their learning, using structured reflection and self-assessment [2], [3].

This paper proposes and evaluates a pedagogical framework for AI-supported mathematics learning that integrates three core elements: (1) prompt engineering as a prerequisite for active engagement, (2) flipped classroom design centred on inquiry-based tasks, and (3) structured reflection and self-assessment to promote metacognitive development.

The contribution of this paper is twofold. First, we provide a concrete and replicable instructional design that operationalises the role of generative AI as a learning partner rather than a solution provider. Second, we present an exploratory evaluation of this framework based on classroom implementation, highlighting how structured reflection and guided AI use influence student learning processes, engagement, and metacognitive awareness.

The study is guided by the following research question: *How does a structured pedagogical framework for AI-supported learning influence students’ engagement, learning strategies, and metacognitive awareness?*

II. RELATED WORK: TEACHER AND STUDENT PERSPECTIVES

Recent research on AI in mathematics education is organised around two main perspectives: the teacher's and the student's.

A. Teacher Perspectives: AI as a Tool for Designing Learning

From the teacher's side, generative AI can enrich differentiated instruction and support professional development [5]. AI can generate practice problems, quizzes, and complete assignments tailored to different ability levels. Cho and Kim [6] analysed AI-based platforms (KnowRe Math, ALEKS) and found that two-way adaptation works best, but none offered flexibility in presentation format, a gap teachers can fill manually.

Holman et al. [7] conducted a systematic review of 14 empirical studies (post-2020) on AI interventions in mathematics education. They found that intelligent tutoring systems (ITS) significantly improve learning outcomes when used over longer periods, and tools like Photomath showed a large effect size ($d = 1.37$) in pre-calculus. However, shorter interventions or those without substantial teacher involvement yielded smaller effects. The KI@school project [5] explicitly focuses on how generative AI can foster mathematical thinking and reflective practice.

Building personalised learning platforms and task banks.

At NTNU in Ålesund, ongoing work involves developing a task bank to differentiate problems based on student responses. The goal is to provide each student with practice problems tailored to their current level of understanding, thereby reducing frustration and promoting mastery. This approach is similar to the multi-agent AI tutoring platform proposed by Chudziak and Kostka [8], which combines adaptive feedback, structured course generation, and textbook-based knowledge retrieval. Such platforms go beyond reactive assistance, offering structured, individualised learning experiences. A systematic review on GenAI and personalisation [9] confirms that the effectiveness of such systems depends on teacher-AI collaboration, data privacy, and equitable access. Ethical and regulatory challenges are further discussed in [10].

B. Student Perspectives: Learning to Use AI as a Learning Partner

From the student's side, the main challenge is to use AI as a training partner rather than a substitute. Zhang and Yang [11] emphasised that without structured teacher guidance on prompting, students risk over-reliance and shallow learning. Nikolic et al. [12] noted that while GenAI shows promise as a supplementary tool, minor inaccuracies are frequently observed. The literature has identified several risks associated with uncritical AI use, including over-dependence, loss of mathematical identity, and the development of a "techno-authoritative conviction" scheme [13]–[15]. Several studies have also documented AI hallucinations as a persistent problem [16]–[18]. Wulff and Kubsch [19] discuss the double-edged sword of GenAI in STEM education. These risks

are analysed in detail with concrete examples in the Discussion section.

III. METHOD

This study follows an exploratory case study design situated in authentic classroom practice with a cohort of engineering students ($N=145$) at HVL. The aim is not to provide statistically generalisable results, but to investigate how a structured pedagogical framework for AI-supported learning functions in practice and how students interact with AI when guided by explicit instructional design. Data sources include student-produced prompts, written reflections, and observational insights from classroom implementation.

The data used in this exploratory study consist primarily of qualitative sources, including student-generated prompts, written reflections, and observational notes from classroom interactions. These data were analysed through an interpretive approach, focusing on identifying recurring patterns in how students engaged with AI and reflected on their own learning processes.

The analysis was conducted using a thematic coding approach, where recurring patterns in student prompts and reflections were identified and categorised. The coding process was iterative and involved multiple passes through the data to refine emerging themes, informed by concepts from self-regulated learning theory.

Our pedagogical approach integrates three components: (1) explicit prompt engineering training, (2) a flipped-classroom investigation using Newton's method and bisection, and (3) structured reflection and self-assessment.

A. Prompt Engineering as a Precondition for Active Use

Students were taught that good prompts elicit guidance, not just answers.

A bad prompt: *Give me the Python code for Newton's method.*
A good prompt: *"I need to compare Newton's method and the bisection method to find the root of $x^2 - \cos(x) = 0$. Can you help me write Python code that computes the solution, counts iterations for each method, and explains why Newton's method converges faster?"* This invites explanation and shows the student's own thinking.

B. Flipped Classroom Case Study

Students were given an assignment to solve an equation using numerical methods (Newton's method vs. bisection) without prior instruction on the methods themselves. The assignment was titled *"Flipped classroom with AI as a learning partner"*. However, they received a short introduction to the assignment itself, learning how to use AI as a successful advisor. The problem was to:

- 1) Explain both methods and the Intermediate Value Theorem.
- 2) Develop Python code for $f(x) = x^2 - \cos(x)$ for tolerances $10^{-2}, 10^{-4}, 10^{-6}$.
- 3) Reflect on strengths and weaknesses of each method.

Students were also given structured reflection questions to guide their post-task analysis. Students documented shortly, then wrote a short reflection.

From a theoretical perspective, this flipped classroom design aligns with constructivist and inquiry-based learning principles, where students actively construct understanding through exploration rather than passive reception. The integration of AI as a dialogic partner extends this model by enabling iterative explanation, immediate feedback, and adaptive support during the learning process.

Recent research suggests that combining flipped classroom approaches with AI tools can enhance student engagement and self-regulated learning, particularly when students are required to actively articulate their reasoning and reflect on feedback [20]. This is consistent with findings from intelligent tutoring systems research, where sustained interaction and feedback are critical for learning gains [7].

However, unlike traditional ITS, generative AI introduces epistemic uncertainty, requiring students to critically evaluate responses rather than accept them as authoritative. As emphasised by Zhang and Yang [11], explicit scaffolding in prompting and evaluation is therefore essential. In this context, the flipped classroom does not merely shift content delivery outside the classroom, but restructures the cognitive processes through which students engage with both mathematics and AI.

C. Reflection Questions for Students

After the investigation, students answered the following structured reflection questions [2]:

Basic understanding:

- What is the fundamental difference between Newton's method and the bisection method?
- Why does Newton's method require the derivative, and what happens if the derivative is zero?
- Explain why the bisection method always converges, while Newton's method may diverge.

Convergence and speed:

- What does "quadratic convergence" mean, and which method exhibits it?
- How does the initial guess affect Newton's method? Give an example where it fails.
- Why does the bisection method converge linearly?

Practical considerations:

- For $x^2 - \cos(x) = 0$, why is $[0, 1]$ suitable for bisection?
- Would $x_0 = 0.5$ be a good starting point for Newton's method? Justify.
- How does tolerance (e.g., 10^{-3} vs 10^{-6}) affect iterations and accuracy?

Metacognition:

- What was the most challenging part? Why?
- How did AI help or hinder your understanding?
- If you had to teach one method to a classmate, how would you explain it?
- What questions do you still have about root-finding or using AI as a learning tool?

D. Self-Assessment Rubric

Students completed the self-assessment shown in Table I to evaluate their mastery and identify areas for further practice.

TABLE I
SELF-ASSESSMENT RUBRIC FOR NUMERICAL METHODS (1=LOW, 4=HIGH)

Skill	1	2	3	4
Explain bisection method	1	2	3	4
Explain Newton's method	1	2	3	4
Choose interval for bisection	1	2	3	4
Choose initial guess for Newton	1	2	3	4
Compute iterations for tolerance	1	2	3	4
Identify Newton failure cases	1	2	3	4
Interpret speed vs. reliability	1	2	3	4
Use AI to verify (not copy)	1	2	3	4

IV. RESULTS

A. Observations from the Case Study

Key observations:

- Students using more elaborated prompts appeared to develop conceptual understanding more effectively than those who primarily copied code.
- Many initially fell into Pitfall 1 (asking only for code). After prompt engineering and reflection, understanding improved.
- Flipped classroom + AI freed class time for conceptual discussion.
- Self-assessment helped students identify gaps (e.g., choosing initial guesses) and target practice.

While these findings are based on qualitative observations and are not statistically generalisable, they provide indicative insights into how structured pedagogical design can influence student interaction with AI and support the development of more active and reflective learning strategies.

These observations can be interpreted through the lens of self-regulated learning. Students who used elaborated prompts demonstrated behaviours associated with planning, monitoring, and evaluation [3]. In contrast, students relying on minimal prompts exhibited surface learning strategies focused on answer retrieval rather than conceptual understanding [2].

The flipped classroom structure appears to amplify these differences, as it creates space for independent exploration prior to formal instruction. In this sense, AI acts as a multiplier of existing learning strategies: it supports deep learning when used reflectively, but reinforces superficial approaches when used uncritically.

B. Student Feedback on Metacognitive Benefits

Based on informal student feedback and reflective responses, several students reported that the reflection questions and self-assessment increased their awareness of their own learning processes, not only in mathematics but also in other subjects. One student said: "The self-assessment made me realise exactly what I need to work on, not just in maths but in physics as well." Another noted: "Reflecting on how I

used AI helped me see that I was just copying answers before. Now I ask better questions in all my courses.” Several students explicitly stated that they transferred this metacognitive habit to other courses, where they began to ask themselves: “What do I actually understand here? What do I need to focus on?”

V. DISCUSSION

The findings from the case study can be understood as an interaction between pedagogical design and AI affordances. The flipped classroom creates conditions for active engagement, while structured reflection transforms AI use into a metacognitive process. This supports the broader conclusion that the effectiveness of AI in education depends less on the technology itself and more on how it is embedded within instructional design [9], [10].

A. The Risk of Becoming Passive Users – Six Major Pitfalls

Despite its potential, uncritical AI use carries serious risks. The six pitfalls presented below are synthesised from both the literature and classroom observations. While prior studies have identified individual risks such as hallucinations, over-reliance, and loss of learner agency [13]–[15], this paper contributes by structuring these risks into a coherent pedagogical framework, each accompanied by concrete examples and actionable prevention strategies grounded in instructional design. This structuring is intended to support both analysis and practical implementation, bridging the gap between theoretical risks identified in the literature and concrete classroom practice.

Pitfall 1: AI as a fact machine – Students ask only for the final answer and copy it without understanding.

Problem: The student simply requests “What is the answer?” and copies the AI’s response. They cannot solve similar problems independently and fail to notice when the AI makes mistakes.

Example: A student asks “Derive $f(x) = x^2 \sin(x)$ ” and receives $f'(x) = 2x \sin(x) + x^2 \cos(x)$. They copy this answer without understanding the product rule. When they later encounter $f(x) = x^3 \cos(x)$, they are completely lost.

Prevention: Teach students to formulate prompts that request guidance and explanation, not just the final answer. Encourage them to ask: “Is my solution correct?” rather than “What is the answer?”. This shifts the interaction from answer-seeking to dialogue-based verification.

Pitfall 2: Uncritical acceptance of AI hallucinations – AI can produce confident but incorrect mathematics. Steinbach et al. [16] found that even 100% hallucinations may still lead to learning gains but with higher confusion. Banerjee et al. [18] documented that LLMs frequently make sign errors and mistakes with the chain rule.

Problem: Students assume “ChatGPT said it, so it must be correct.” They do not question AI outputs, even when the AI invents plausible-sounding but false information.

Example: A student asks “Derive $f(x) = e^{2x}$ ”. The AI responds $f'(x) = e^{2x}$ (forgetting the chain rule). The correct answer is $f'(x) = 2e^{2x}$, but the student copies the wrong answer without checking.

Prevention: Deliberately show AI errors in class. Teach cross-checking using textbooks or formularies.

Pitfall 3: Passive learning and skill atrophy – Students conclude: “I don’t need to practice – AI will do it for me.”

Problem: The brain is not trained. Mathematical skills do not develop. There is no deep learning or lasting understanding. Students cannot solve problems without AI assistance, for example on proctored exams.

Example: A student says “I understand when I see the solution, but I can’t do it myself” or “Without AI, I cannot derive $f(x) = \ln(3x^2)$.”

Prevention: Use proctored assessments and reflective prompts. Mandate that certain core tasks be done without AI.

Pitfall 4: Over-trust and techno-authoritative conviction – Students treat AI as an unquestionable authority [13].

Problem: Students develop a “techno-authoritative conviction” scheme where AI outputs are accepted as truth without critical evaluation. This erodes their ability to think independently and judge mathematical correctness.

Example: When asked “Why is this correct?” the student responds “Because ChatGPT said so,” rather than providing a mathematical justification based on definitions or theorems.

Prevention: Train students to identify errors. Use “adversarial examples” where the AI deliberately provides wrong answers for students to catch [17].

Pitfall 5: Erosion of foundational skills (brittle knowledge) – Skipping basic skill-building leads to a “brittle” knowledge base, as also discussed in institutional guidelines [15].

Problem: Students bypass fundamental practice, believing AI will always be available. This results in knowledge that collapses under pressure or when AI is unavailable. Foundational skills like algebraic manipulation, differentiation rules, and integration techniques are never internalized.

Example: A student uses AI to solve all derivative problems for a chapter on the product rule. When faced with a combined problem requiring both the product rule and the chain rule, they cannot break it down because they never mastered the individual components.

Prevention: Mandate certain core tasks be completed without AI. Design assessments that specifically test foundational skills.

Pitfall 6: Loss of mathematical identity and agency – Over-reliance erodes students’ sense of themselves as capable thinkers [14].

Problem: Students begin to see mathematics as something only AI can do. They lose confidence in their own abilities and develop a fixed mindset about mathematical talent. This undermines their willingness to struggle productively with challenging problems.

Example: A student says “I’m just not good at math” or “Math is for computers anyway.” They no longer see themselves as someone who can think mathematically or solve problems creatively.

Prevention: Normalise struggle as part of learning. Use reflective journals where students document their problem-solving process. Emphasise that AI is a training partner, not a substitute.

B. Student Checklist for Responsible AI Use

Based on the six pitfalls above, we provide students with the following checklist to promote responsible AI use:

- 1) Have I tried the problem myself first?
- 2) Have I written down my own reasoning?
- 3) Am I asking for guidance, not just the final answer?
- 4) Have I critically evaluated the AI's response (check with textbook or formula collection)?
- 5) Can I solve a similar problem without AI afterwards?
- 6) Have I reflected on what the AI did well and where it went wrong?

C. Implications for Teaching

The results show that structured reflection and self-assessment are essential for preventing these six pitfalls. When students regularly articulate their understanding, identify gaps, and evaluate progress, they build metacognitive skills that transfer beyond mathematics. Self-assessment moves students from passive receipt of AI-generated answers to active monitoring of their own competence. Reflection forces them to compare methods, justify choices, and consider real-world applicability, precisely the deep learning outcomes that survive in an AI-rich environment.

Student feedback confirms that these practices have increased awareness of learning processes across subjects. We therefore recommend that any AI integration in mathematics education be accompanied by systematic opportunities for reflection and self-assessment. These practices restore student agency and mathematical identity, countering over-dependence and passive consumption.

D. Limitations and Future Work

Our findings are based on a single case study at two institutions. Future work will evaluate a multi-agent AI tutoring platform with adaptive feedback and a task bank for personalised problem generation [8]. We also plan to investigate long-term effects on mathematical identity and self-efficacy. In the upcoming semester, we intend to conduct further empirical investigations to systematically examine how the three core pedagogical components proposed in this article, prompt engineering, flipped classroom design, and structured reflection with self-assessment, can be implemented more broadly and how they influence student learning outcomes across different mathematical topics and student populations.

VI. CONCLUSION

Artificial intelligence is not a temporary trend but a permanent fixture in the educational landscape. Attempting to ban or restrict student access to AI is both impractical and pedagogically misguided. Instead, we must teach students to use AI as a *training partner*, a tool that supports, but never

replaces, their own thinking. This shift requires a fundamental change in the roles of both students and teachers.

From the Student's Perspective

Students must move from passive consumers of AI-generated answers to active, self-regulated learners. This means:

- **Attempting first:** Always trying to solve a problem on their own before consulting AI.
- **Prompting with purpose:** Asking for explanations, verification, or alternative approaches, not just final answers.
- **Reflecting on outcomes:** Using checklists and self-assessment questions to evaluate what they have learned and what remains unclear.
- **Practicing without AI:** Regularly testing themselves in AI-free environments to ensure genuine skill development.

When students internalise these habits, AI becomes a catalyst for deeper understanding rather than a shortcut.

From the Teacher's Perspective

The teacher's role evolves from being the sole source of answers to a designer of learning experiences that require uniquely human cognitive work. Key responsibilities include:

- **Designing reflective tasks:** Assignments that ask students to compare, critique, and justify AI outputs, not just compute.
- **Teaching AI literacy explicitly:** Demonstrating good prompting, pointing out common AI errors, and discussing ethical use.
- **Creating formative self-assessment tools:** Checklists, rubrics, and reflection prompts that help students monitor their own learning.
- **Maintaining proctored assessments:** Ensuring that some portion of evaluation remains AI-free to verify individual competency.

By embedding these practices into everyday teaching, the teacher transforms the classroom into a community of reflective, critical thinkers.

The Pedagogical Core

Our case study with Newton's method and the bisection method provides indications of improved conceptual understanding and increased problem-solving confidence. The success of this approach hinges on three pedagogical pillars:

- 1) **Good prompting:** teaching students how to ask questions that elicit explanation, not just answers.
- 2) **Reflection and self-assessment:** requiring students to articulate what they learned, what they still struggle with, and how they plan to improve.
- 3) **Formative tasks with critical evaluation:** designing assignments where students must compare, verify, and critique AI-generated solutions.

The Ultimate Goal

The purpose of this pedagogical approach is not to restrict AI use, but to facilitate good use of AI that contributes to deep learning. We want students to understand mathematics deeply enough to use AI as a partner in exploration, a checker of routine work, and a generator of alternative perspectives. At the same time, AI can be used in multiple ways to increase learning outcomes and streamline guidance, for example, by providing instant feedback on routine tasks, generating personalized practice problems, and freeing teacher time for more meaningful interactions.

Reflection, self-assessment, well-designed control questions, and formative tasks that require students to apply, compare, and justify – these are the pedagogical keys to achieving that goal. By embedding these practices into everyday teaching, we transform mathematics education into a journey of human intellectual growth, preparing students not just for exams, but for a lifetime of thoughtful, responsible problem-solving in an AI-augmented world. Graduates who emerge from such programmes are prepared to engage with AI critically and constructively, using it as a supportive tool while preserving their own judgment, creativity, and ethical responsibility.

These patterns suggest that the quality of student-AI interaction is a key mediating factor in learning outcomes, rather than the mere presence of AI tools.

TABLE II
KEY RECOMMENDATIONS FOR INTEGRATING AI IN MATHEMATICS EDUCATION

1. Teach AI literacy: good prompting (ask for explanation, not just answers), error detection, ethical use.
2. Require students to attempt problems first, then use AI for verification and reflection.
3. Use AI for formative feedback (e.g., checking routine work), but maintain proctored, AI-free summative assessments.
4. Leverage AI for your own teaching: generate practice problems, analyse student misconceptions, differentiate tasks.
5. Create clear institutional policies on permitted AI use and documentation (as discussed in [15]).
6. Address the six pitfalls explicitly in teaching: fact machine, hallucinations, passive learning, over-trust, brittle knowledge, loss of identity.
7. Integrate structured self-assessment, reflection questions, and control questions into every AI-augmented activity.

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